One last thing.

Does $\lim_{r \to 1} u(r,\theta) = f(\theta)$ i.e. is $u \in C^0(\bar{\Omega})$

Not responsible for ~~We need~~

We can show this directly, but we need an integral representation of $u(r,\theta)$ to show this.

Subst def'n of a_n, b_n into our sol'n:

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\phi)\,d\phi + \sum_{n=1}^{\infty} r^n \frac{1}{\pi}\int_{-\pi}^{\pi}\Big[\cos(n\theta)\cos(n\phi) + \sin(n\theta)\sin(n\phi)\Big]\,d\phi.$$

$$[\ldots] = \cos[n(\theta-\phi)]$$

Since series converges uniformly, for $r<1$, we can interchange Σ & $\int$

check on $\frac{1}{2\pi}$

ok

$$u(r,\theta) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\phi)\left[1 + 2\sum_{n=1}^{\infty} r^n \cos[n(\theta-\phi)]\right] d\phi$$

No $\frac{1}{2\pi}$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(\phi)\, P(r, \theta-\phi)\, d\phi$$

where $P(r,\xi) = \frac{1}{2\pi}\left[1 + 2\sum_{n=1}^{\infty} r^n \cos(n\xi)\right]$

Uses $z = re^{i\xi}$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

$$= \frac{1}{2\pi}\,\frac{1-r^2}{1+r^2-2r\cos(\xi)}$$

$$u(r,\theta) = \begin{cases} \frac{1}{2\pi}\int_0^{2\pi} f(\phi)\,\frac{1-r^2}{1+r^2-2r\cos[n(\theta-\phi)]}\,d\phi & r<1 \\ f(\theta) & r=1 \end{cases}$$

Using this representation we can show

$$\lim_{r\to 1} u(r,\theta) = f(\theta)$$

(details of this argument for 3D is given in §7.9)

$P(r,\xi)$ = Poisson kernel

= an example of Green's f'n.

So we have a sol'n to the Dirchlet problem for the Laplace eq'n on unit disk

① F-S sol'n
② using Poisson kernel

For a disk of radius a $\quad 0<r'<1 \quad 0<r<a$

$(r,\theta) \leftarrow (r',\theta') \qquad r' = \frac{r}{a} \quad \theta = \theta'$

$$u(r,\theta) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left[a_n \left(\frac{r}{a}\right)^n \cos(n\theta) + b_n \left(\frac{r}{a}\right)^n \sin(n\theta)\right]$$

still the avg of $f(\theta)$ over $[-\pi,\pi]$
(just replace r by $\frac{r}{a}$).

Ch 7 section 16 Neumann Problem

$\Omega \subset \mathbb{R}^n$,

$\partial\Omega$ is nice (can apply divergence Thm)

Neumann problem

Find $u \in C^1(\bar{\Omega}) \cap C^2(\Omega)$ so That

$$\Delta u = 0 \quad \text{on } \Omega$$

$$\frac{\partial u}{\partial n} = f \quad \text{on } \partial\Omega$$

where $f \in C^0(\partial\Omega)$ is given.

Uniqueness: None If u is a sol'n, Then $u + C$ is also a sol'n. But we can show:

$$\boxed{\text{Thm: Any 2 sol'ns of The Neumann problem differ by a constant}}$$

<u>Pf</u> Let $u_1, u_2 \in C^1(\bar{\Omega}) \cap C^2(\Omega)$ satisfy the N.P. Let $\tilde{u} = u_1 - u_2$

Then $\Delta \tilde{u} = 0$ because $\Delta u_1 = 0$ $\Delta u_2 = 0$

$\frac{\partial \tilde{u}}{\partial n} = 0$ $\frac{\partial u_1}{\partial n} = f$ $\frac{\partial u_2}{\partial n} = f$

Use Green's 1st identity

$$\int_\Omega u \underbrace{\Delta w}_{0} \, dv = \int_{\partial\Omega} u \underbrace{\frac{\partial w}{\partial n}}_{0} \, d\sigma - \int_\Omega \underline{\nabla} u \cdot \underline{\nabla} w \, dv$$

Let $u = w = \tilde{u}$

$$0 = 0 - \int_\Omega \underbrace{\underline{\nabla} \tilde{u} \cdot \underline{\nabla} \tilde{u}}_{\geq 0} \, dv$$

$$\Rightarrow |\underline{\nabla} \tilde{u}|^2 = 0 \Rightarrow \underline{\nabla} \tilde{u} = 0 \underset{HW}{\Longrightarrow} \tilde{u} = \text{const.}$$